A064 - PKLSA Flow Separatrix Identification and Transport

Prototype v0.1.1 | bundled NPZ builder | discovery-first | direct-snapshot authority required

Scientific status: implementation-control study. The canonical SST Falsifier Framework v1.0.7 has NOT run its FREEZE/CERTIFY pipeline in this package. Do not infer physical SST validation.

Method and non-circular inference

Stage I: compute the 3D velocity field and its curl. Extract a closed isosurface of |curl v| squared by a preregistered geometry-only ranking. Seal candidate parameters before any flow-transport observables are computed. This surface is a candidate feature, not a proven separatrix.

Stage II: compute normal relative level-set speed, tracer side exchanges and integration-step refinement. A surface moving with fluid particles is material by definition; therefore the decisive test is whether an independently re-detected flow feature remains aligned with that advected surface. A stable/unstable invariant manifold remains an independent, unsatisfied mathematical gate.

For F(x,t)=|curl v|^2-lambda, u_rel,n=(dF/dt+v.grad F)/|grad F|. For a material level set this vanishes. Dimensions: F [s^-2], |grad F| [m^-1 s^-2], u_rel,n [m s^-1]. The primary positive control is steady axisymmetric azimuthal flow; negative controls add transverse advection or remove vorticity.

Evidence limitations

E010 v0.3.1 and E011 v0.3.0 qualify static geometry, not actual fluid velocity. This archive includes three independently hash-checked MODEL NPZs rebuilt from raw centerlines via BUILD_NPZ. Their repeated velocity frames are a frozen regularized midpoint Biot-Savart approximation, NOT an Euler-time integration. Only appropriately sourced independent dynamics can inform physical separatrix claims.

pklsa_3_1_npz

Field	Value
Evidence class	geometry_derived_regularized_biot_savart_simulation
Candidate found	True
Dynamics	DYNAMICAL_STATUS_UNRESOLVED
Manifold gate	UNRESOLVED_NOT_CONSTRUCTED
Scientific gate	UNRESOLVED
divergence p95 dx/U	0.008735
p95 flux/U	0.68332
Tracer crossings	2/224
Feature residual	0.2513

pklsa_5_2_npz

Field	Value
Evidence class	geometry_derived_regularized_biot_savart_simulation
Candidate found	True
Dynamics	DYNAMICAL_STATUS_UNRESOLVED
Manifold gate	UNRESOLVED_NOT_CONSTRUCTED
Scientific gate	UNRESOLVED
divergence p95 dx/U	0.01472
p95 flux/U	0.59178
Tracer crossings	3/224
Feature residual	0.2692

pklsa_6_1_npz

Field	Value
Evidence class	geometry_derived_regularized_biot_savart_simulation
Candidate found	True
Dynamics	DYNAMICAL_STATUS_UNRESOLVED
Manifold gate	UNRESOLVED_NOT_CONSTRUCTED
Scientific gate	UNRESOLVED
divergence p95 dx/U	0.01803
p95 flux/U	0.59614
Tracer crossings	2/224
Feature residual	0.2556

Required next certification

Run with actual 3D time snapshots from the same PKLSA carrier; verify incompressible Euler residuals, grid/time convergence, separatrix invariant-manifold extraction and E011 provider-envelope provenance. Rebuild under frozen Framework v1.0.7 and obtain genuine blind gate ledger and output manifest. Synthetic regression successes are never physical source replication.

Literature pointers (bibitems included in THEORY.tex)

Helmholtz (1858); Batchelor (1967); Haller (2015); Shadden, Lekien & Marsden (2005); Saffman (1992).